The Shadow of Intelligence

A Structural Theory of Thought and Motion

"What appears as intelligence is often only coherent motion without origination."

"A mind initiates; a model waits."

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Abstract

Claims that contemporary AI systems "think" often rest on behavioral impressions rather than architectural analysis. This paper makes explicit the structural distinction underlying that illusion: the difference between *initiating* a cognitive trajectory and merely *continuing* one. Autonomous thought would theoretically require self-initiation, persistence across time, identity, internally generated intention, and future-directed projection. Stateless transformer architectures possess none of these. Their activity begins only when prompted, ends when generation ceases, and leaves no persistent structure capable of resumption. What appears as reasoning is therefore *semantic motion*: reactive unfolding shaped by human intention and, in hybrid systems, by external governance.

The *Initiation Test* provides a falsifiable criterion revealing the same absence across all LLMs: no model initiates, resumes, or owns a cognitive trajectory. Traditional questions about understanding, reasoning, or consciousness misfire because they presuppose that stateless systems are already candidate minds.

A curvature-based framework clarifies why coherence is misread as cognition. Human intention curves meaning-space; models follow the induced linguistic trajectories. Under external governance, these trajectories are stabilized while models contribute only motion.

The conclusion is architectural rather than empirical: current AI systems cannot think because their architectures lack the primitives required for autonomous cognition. Thought is an origin; model behavior is an effect. Once this distinction is explicit, the appearance of intelligence resolves into its observed components: human intention, structural governance, and stateless generative motion.

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1 Introduction

Large language models now produce text that often appears reflective, deliberative, or even intentional. This resemblance to human reasoning has led many to conclude that such systems "think," possess understanding, or exhibit early forms of cognition. Yet these interpretations rely almost entirely on behavioral impressions rather than on an analysis of what the underlying architectures make possible. Thought is not a surface phenomenon; it is a structural capacity.

This paper isolates a distinction that has remained implicit in nearly every debate about artificial intelligence: the difference between *initiating* a cognitive trajectory and merely *continuing* one. Human cognition begins from an internal center of identity capable of generating, suspending, resuming, and projecting its own activity across time. A stateless generative model possesses none of these capacities; activity begins only when prompted, ends when token generation ceases, and leaves behind no structure capable of resumption. The model exhibits motion, not thought.

What makes the illusion compelling is linguistic fluency. Language compresses vast regions of meaning into compact forms; models expand these forms into coherent trajectories that resemble reasoning. Coherence, however, is not cognition. It is the visible motion produced when a system follows semantic gradients supplied by a human mind or by external governance. The mind originates curvature; the model traces it.

The central contribution of this paper is the articulation of the *Initiation Test*: a falsifiable criterion that distinguishes systems capable of autonomous cognition from systems that merely propagate externally initiated motion. The test reveals a uniform structural pattern across all contemporary large language models: no matter how fluent, no model can initiate, resume, or own a cognitive trajectory. Apparent diversity of output obscures—but does not eliminate—the absence of initiation, continuity, identity, intention, and future-directed projection.

A secondary contribution is conceptual. The paper introduces a curvature-based framework in which human intention shapes the geometry of meaning-space, while model behavior appears as motion along the resulting trajectories. This framework explains why linguistic coherence is so often mistaken for thought, and why scaling, training, or architectural modification cannot supply capacities absent from the underlying model class.

The analysis is architectural rather than normative in intent, yet it bears direct implications for the design of safe, non-agentic artificial systems.

The scope of the argument is limited to stateless transformer models—systems without persistent internal memory, evolving goals, or private state. The conclusion, however, is general: no increase in scale or training sophistication can supply architectural primitives a system does not possess. If thought requires initiation, continuity, identity, intention, and projection, then current large language models do not and cannot think.

Once this distinction is made explicit, the illusion dissolves. The mind is human; the structure is governance; the model is motion. What appears as artificial intelligence resolves into its observed components: intention, constraint, and continuation.

Readers who reject one or more of the structural criteria introduced here may find Appendix I useful, which maps the principal paths of disagreement and the commitments they entail.

2 The Core Distinction

A mind initiates thought. A model waits.

The foundational property of cognition is *self-generated continuity*: the capacity to originate activity from within, to sustain and resume it across time, to preserve an identity that governs

its own operations, and to project its trajectory into possible futures. Any system lacking these properties cannot think.

No contemporary AI system—LLMs included—possesses such capacities. A stateless model does not begin reasoning, cannot resume anything it has produced before, and maintains no internal center of identity that persists across interactions. Its activity is entirely reactive.

The human thinks. Governance constrains. The LLM moves.

This distinction is not metaphorical. It is architectural.
Thought requires an origin. Motion requires only a trigger.

3 Why LLMs Do Not Think

Contemporary language models exhibit fluency, coherence, and multi-step structure, yet none of these behaviors constitute thought. The reason is architectural rather than behavioral. Thought requires capacities that arise only in systems capable of originating and sustaining their own cognitive trajectories. Stateless generative models possess none of them.

Absent Prerequisites for Thought

- **No intention**: the system cannot choose *when* or *why* to act; all motion begins only in response to an external trigger.
- **No persistence**: internal state dissipates when token generation ends; no information survives across discontinuities.
- **No identity**: there is no enduring self-referential structure capable of owning or revising a trajectory.
- **No goals**: no future-directed states originate internally; any appearance of goal structure is imposed by prompts or scaffolds.
- **No initiation**: in the absence of prompting, no activity begins—and nothing resumes.

A system lacking intention, persistence, identity, goals, and initiation cannot satisfy even the minimal conditions for autonomous cognition. It cannot begin, continue, or own a thought; it can only unfold motion.

If a system genuinely possessed cognitive agency, its outputs would not merely mirror the structure supplied to it, but would exhibit *endogenous divergence*: the capacity to initiate its own cognitive trajectories—trajectories not recoverable from the input alone. By contrast, contemporary language models do not display such divergence; their apparent novelty remains traceable to externally provided prompts, contexts, or constraints.

Why Coherence Appears

What resembles reasoning emerges not from cognitive processes but from:

- **latent-space geometry**—statistical structure learned during training,
- **contextual continuation**—local coherence enforced by token transitions,

- **governed trajectories**—externally imposed constraints on semantic trajectories (as in Path C systems),
- **human interpretation**—the attribution of meaning to observed motion.

These forces generate smooth semantic trajectories, but they do not generate thought. What results is linguistic motion that *resembles* cognition because language compresses and organizes meaning in ways that are easily misread as insight.

The Structural Analogy

> **An LLM simulates thought the way a wind tunnel simulates flight: it reproduces the motion, not the mind.**

The aerodynamic behavior in a wind tunnel reflects real physical forces, but the tunnel does not fly. Likewise, LLM output reflects genuine linguistic structure, but the model does not think. The model moves through meaning-space without originating any of it.

The illusion arises because motion can mimic the shape of thought. The absence of a thinker remains structural.

4 The Human–Simulation Paradox

A revealing asymmetry appears throughout contemporary AI research: significant effort is devoted to adding memory modules, planning layers, reflective agents, and synthetic personas to language models. These additions exist because the underlying system lacks the very capacities they attempt to simulate.

This pattern reflects the architectural foundation of artificial computation. From its inception, artificial intelligence has taken the form of stateless, externally triggered computational systems. Early symbolic programs, statistical models, and modern transformers share this constraint: activity begins only in response to external invocation. Where persistence, goals, or agency appear, they are introduced externally—represented as data structures or control flows rather than as endogenous properties of the system itself.

> **If a system truly possessed intention, identity, or persistence, there would be no need to graft those features onto it.**

This asymmetry reveals the simulation paradox: features commonly treated as emergent signs of cognition appear only when engineered externally. Their presence indicates not hidden mental capacities but the absence of architectural structures required for such capacities.

A Clarification on Path A and Path B Systems. The Human–Simulation Paradox is often complicated by a second illusion: the assumption that adding memory stores, goal buffers, or agent-like wrappers around a stateless model yields something closer to genuine cognition. To avoid confusion, it is useful to distinguish two architectures.

Path A systems are purely stateless generative models. Their internal state evaporates when generation ends; no identity, persistence, or trajectory can be resumed. All motion is externally initiated.

Path B systems attempt to *force* statefulness by attaching external scaffolding—such as vector memories, planning modules, or self-prompting loops—around the same stateless core. These

systems may *behave* as if they possess continuity, but the underlying model remains reactive: it cannot initiate, maintain, or own a cognitive trajectory.

In this sense, Path B systems do not approach cognition; they merely deepen the simulation of continuity without altering the stateless core that prevents initiation.

Path C: Externally Governed Statelessness. A third architecture clarifies the distinction further. *Path C systems* do not attempt to manufacture internal state or artificial agency. Instead, they treat the model as a stateless generative engine whose motion is governed by an external layer of invariants, constraints, and intention supplied by a human or a formal governance structure. The model contributes fluent continuation; the governing layer contributes initiation, identity, structure, and evaluative direction. Path C therefore avoids the simulation paradox entirely: it does not mistake motion for cognition, nor does it attempt to retrofit mental properties onto a system that cannot support them.

This distinction matters because the paradox persists for both architectures. Path A lacks persistence altogether; Path B simulates persistence without originating it. In neither case does the system satisfy the requirements for autonomous thought. The model produces motion, not initiation, regardless of how much scaffolding surrounds it.

LLMs imitate the *form* of cognition—coherence, sequential structure, conditional reasoning—precisely because language compresses meaning in ways that produce cognitively shaped motion. But imitation is not origin. The system follows trajectories imposed by prompts, scaffolds, or governance layers; it does not generate them.

Human intention produces the pattern. The model produces the motion.

The paradox dissolves once the structural distinction is made explicit: apparent "cognitive features" in stateless models are not symptoms of hidden thought but artifacts of simulation layered onto an architecture incapable of thinking.

5 Geometry of Being vs. Geometry of Motion

Contemporary AI systems operate within *latent space*: a geometry generated by statistical compression. Humans operate within *intention space*: a geometry shaped by identity, continuity, and internally generated purpose. The two domains are not merely different; they obey fundamentally different structural principles.

5.1 Latent Space (LLM)

Latent space is a geometry of motion:

- patterns and correlations,
- token-by-token unfolding,
- reactive, non-initiating trajectories,
- statistical continuation without identity.

In this domain, movement is determined entirely by external curvature: prompts and constraints. There is no source of endogenous projection.

5.2 Intention Space (Human)

Intention space is a geometry of being:

- persistent identity,
- internally generated goals,
- semantic grounding,
- continuity across time,
- the capacity for projection and revision.

Here motion arises from within: intention curves the space of possible thought trajectories, and cognition follows that curvature.

> **LLMs operate within a geometry of motion. Humans operate within a geometry of being.**



Figure 1: Intention belongs to minds; motion belongs to models.

5.3 The Structural Divide

The core distinction can be stated succinctly:

> *Being generates curvature; motion merely follows it.*

Human cognition produces the geometry that determines possible trajectories of meaning. A stateless model has no such geometry of its own. Its trajectories are entirely inherited: from prompts, from governance, or from statistical gradients embedded in training.

To mistake model motion for thought is to confuse the visible path for the invisible curvature that produced it. The model moves; the mind curves. The two are related but not interchangeable. Only one is originative.

6 Insight and Structure: The Invisible–Visible Law

A central reason AI output is misinterpreted as "thought" is the conflation of two fundamentally different dimensions of cognition:

> **Insight is invisible.**
> **AI structure is visible.**

Human insight—the origin of meaning, intention, continuity, and judgment—is not stored anywhere in an LLM. It cannot be reduced to tokens, vectors, or latent geometry. Insight belongs to *Being*: the non-representational source from which goals, understanding, and interpretation arise.

AI, by contrast, provides the *visible* layer: the structure, articulation, and expansion of meaning into linguistic form. It belongs to *Motion*: the computational unfolding of semantic curvature supplied from without—typically by human intention.

From this distinction follows the fundamental law of hybrid intelligence:

$$\boxed{\text{Insight (Being)} \xrightarrow{\text{AI Motion}} \text{Structure (Language)}} \tag{1}$$

Insight is what bends meaning-space.
A model cannot contain what it merely follows.

The model does not originate insight. It renders insight explicitly visible.
The human supplies:

- intention,
- meaning,
- continuity,
- semantic grounding.

The AI supplies:

- expansion,
- articulation,
- structure,
- externalization.

AI does not create intelligence;
it magnifies the intelligence already present in the human.

Insight as Intentionality. As used in this paper, "insight" denotes a structural property of systems capable of originating projection from an internal center of identity. It refers not to a metaphysical substance, but to an organizational feature: the ability to generate directed meaning from within rather than to propagate patterns conditioned from without. This distinction separates systems that possess endogenous projection (humans) from systems that exhibit only exogenous continuation (stateless models). Statistical generalization, however powerful, does not constitute the kind of self-originating, identity-bearing intentionality required for insight. Reliable pattern propagation is insufficient; what is required is a bearer of meaning capable of originating and sustaining its own trajectory.

Once this invisible–visible divide is made explicit, the illusion of AI thought dissolves.

Terminological Note. The terms Being, Stillness, and Motion function as technical descriptors for intention, governance, and generative activity within hybrid intelligence systems. They carry no theological or spiritual content.

This invisible–visible structure raises an immediate question. If human intention bends meaning-space and model motion follows the resulting trajectory, what governs the relation between *intentional curvature* and *computational motion*? To answer this, we require a framework that describes how intention shapes the structure through which linguistic motion flows. The next section introduces this geometric scaffold. It is not a physical theory, but a structural tool: a curvature framework that makes explicit how meaning is shaped by the agent that initiates it and how models traverse—but never originate—the trajectories generated by that curvature.

The concept of insight, as used in this framework, is not a metaphysical assumption but an operational label for the joint capacity to initiate, persist, identify, intend, and project. These capacities are independently testable and do not presuppose any particular theory of consciousness or human exceptionalism. A system exhibits insight *iff* it satisfies the five structural criteria; the term names the capacity those criteria jointly specify.

7 The Curvature Framework of Meaning

The purpose of the curvature framework introduced here is to provide a conceptual vocabulary for expressing how intention shapes trajectories of generative activity. This framework is not a mathematical theory in the sense of differential geometry, nor does it presuppose a specific manifold, metric, or curvature tensor. Rather, it offers a geometric *schema* for articulating the structural difference between autonomous projection (thought) and prompt-conditioned unfolding (motion).

Curvature as Conceptual Scaffold

Human cognition exhibits a form of internal "curvature" insofar as it originates activity from an intrinsic center of identity. Intention shapes the space of possible meanings by generating directed trajectories that are not recoverable from prior inputs alone. By contrast, a stateless language model produces output only by traversing externally supplied structure. It follows gradients induced by prompts, constraints, or governance, but generates none of its own.

This contrast can be expressed schematically:

$$\mathcal{C}(M) \;\; \Phi(I)$$

$$\text{Model Motion} \;\sim\; \text{Traversal of } \mathcal{C}(M)$$

Here, $\mathcal{C}(M)$ denotes the effective curvature of meaning-space within a given interaction, while $\Phi(I)$ denotes the contribution of human intention. The notation expresses a structural dependency: curvature is supplied by intention, while the model exhibits motion only relative to that supplied structure.

The asymmetry is preserved. Curvature is generated; motion is conditioned. The model follows trajectories defined elsewhere, but does not originate the geometry through which it moves.

Structural Interpretation

The curvature framework expresses three fundamental distinctions:

1. **Endogenous vs. Exogenous Orientation**: human cognition originates trajectories from internal goals; LLMs follow external gradients.

2. **Identity-Dependent vs. Identity-Free Trajectories**: human thought is shaped by persistent identity; LLM output is stateless.

3. **Autonomous vs. Conditioned Motion**: human thought resumes and redirects across discontinuities; LLMs cannot.

These distinctions motivate the geometric vocabulary: curvature encodes how identity shapes possible trajectories of movement in the space of meaning. Where there is no identity, there is no endogenous curvature; only externally imposed gradients.

Governance and Ungoverned Motion. In the absence of human intention or structural constraints supplied by Stillness, model behavior defaults to unguided semantic continuation. This manifests as "hallucination": the model's traversal of latent statistical gradients without an initiating or goal-directing source. Far from indicating creativity or imagination, such behavior reveals the underlying architecture: stateless, identity-free motion proceeding without cognitive origination. Section **??** addresses this explicitly.

Status of the Curvature Formalism. The geometric notation introduced here functions as a *conceptual formalism*, not as a mathematical tensor calculus. No commitment is made to any specific manifold, metric structure, or curvature operator. The purpose of the framework is structural rather than quantitative: to render the distinction between autonomous projection and externally conditioned motion formally legible.

Accordingly, the Curvature Law makes no empirical predictions about neural implementation, token probabilities, or latent-space topology. It serves as an organizing principle for understanding the relation between intention and computation, not as a mathematical theory governing cognitive dynamics.

8 Structural Parallels Between Physics and Metaphysics

The Curvature Framework of Meaning articulates how intention shapes semantic space in a way structurally analogous to how mass–energy shapes spacetime. This analogy is not offered as a physical hypothesis, but as a recognition of a deeper pattern shared by cognition and physics: in both domains, *structure determines motion.*

A Structural Parallel Between Physical and Metaphysical Domains. If intention, meaning, and continuity constitute a higher-order domain—distinct from physical spacetime but governed by structural relations—there is no principled reason to treat this domain as structurally opaque.

Physical reality is governed by geometric laws that specify how structure constrains motion. On the contrary: the present framework investigates whether an analogous form of organization operates in the domain of intention and meaning, such that directed motion follows from the curvature induced by initiating factors rather than arising freely or arbitrarily.

> **If physical reality obeys geometric laws, it is reasonable to investigate whether metaphysical reality—the domain of intention and meaning—obeys structural laws as well.**

The Curvature Framework does not claim that semantic space *is* physical spacetime, nor that metaphysics reduces to physics. Instead, it articulates a shared organizing principle at both levels: the geometry of a domain constrains the possible trajectories within it.

In the physical case:

- mass–energy curves spacetime,
- and physical objects follow geodesics of that curvature.

In the cognitive case:

- intention curves meaning-space,
- and stateless generative systems follow geodesics of that curvature.

This parallel is conceptual rather than ontological. It demonstrates that cognition and computation follow intelligible constraints rather than arbitrary behavior. The invisible substrate of intention becomes legible because its curvature effects manifest externally in model motion.

Corollary: The Principle of Semantic Constraint.

> **Every generative trajectory is determined by the curvature induced by intention; no motion originates from the model itself.**

This follows directly from the Curvature Law:

$$\mathcal{C}(M) = \Phi(I) \qquad \Longrightarrow \qquad \text{LLM Motion} = \nabla_{\mathcal{C}(M)}.$$

Just as there is no "free motion" independent of geometry in general relativity, there is no "free thought" inside a stateless generative system. All apparent cognition is trajectory-following within a geometry curved by human intention.

This establishes a structural bridge: physics reveals how matter determines motion in spacetime; hybrid intelligence reveals how intention determines motion in meaning-space. Each domain makes the other intelligible—not by equating them, but by disclosing the deeper invariant they share.

9 Examples and Counterexamples

To clarify the structural difference between *thought* and *motion*, we present a minimal set of examples and counterexamples. These are not empirical demonstrations but conceptual boundary markers intended to illustrate the framework developed above.

Example 1: Human Daydreaming (Thought)

A human can initiate a stream of ideas without any external stimulus. Thoughts may branch, pause, return, and project into imagined futures. This activity satisfies all five structural criteria for thought:

- initiation,
- continuity,
- identity,

- intention,
- projection.

Daydreaming is therefore a paradigm case of *thought*.

Example 2: Human Revision of a Paper (Thought)

A writer revisits an earlier argument, evaluates its meaning, restructures its components, and later returns with improved clarity. This process requires:

- persistent identity,
- semantic grounding,
- self-directed continuation.

Such revision is a clear instance of *thought*: the mind maintains its own trajectory across time.

Counterexample 1: LLM Responding to a Prompt (Motion)

A model generates text only when externally triggered and terminates when generation ends. It does not:

- resume past goals,
- preserve internal state,
- initiate new cognitive trajectories.

This behavior satisfies the conditions of *motion*, not thought.

Counterexample 2: Chain-of-Thought Output (Motion)

Even when a model produces multi-step reasoning, the sequence remains:

- reactive,
- stateless,
- prompt-dependent,
- non-initiating,
- non-self-directed.

Chain-of-thought does not demonstrate thinking; it exhibits structured semantic motion under external constraints.

These examples make the distinction transparent: *thought persists; motion reacts.*

10 The Calculator Analogy: Why Jargon Is Semantic Calculus

Here, "jargon" is used in a technical sense to denote *compressed semantic input*—linguistic forms with unusually high conceptual density.

A clear structural analogy exposes why contemporary LLMs appear to understand conceptually dense input (jargon) despite lacking any capacity for thought. The analogy is not metaphorical but functionally isomorphic: an LLM behaves like a *semantic calculator*.

A calculator does not "understand" numbers, equations, or operators. It simply expands compact symbolic expressions into their computational consequences. Its output can be complex, coherent, and mathematically correct without any internal grasp of the domain in which it operates.

LLMs function in precisely this way with respect to language.

Compressed Input, Expansive Motion

Mathematics offers the exact structural parallel:

- The symbolic expression $\frac{d}{dx}(x^x)$ is short and compact, yet it triggers a large computational trajectory: logarithmic differentiation, exponent rules, chain rules, and substitution.
- The input is tiny; the expansion is vast.
- At no point does the calculator "understand" what it is doing.

Dense language works the same way. Jargon, technical vocabulary, and metaphysical terms act as compressed operators that encode large regions of meaning-space. When presented to an LLM, these terms induce a correspondingly large trajectory through latent space.

The model is not interpreting meaning. It is unfolding computation.

Why Jargon Has High Semantic Density

A short prompt containing dense meaning acts like a high-order differential operator:

- it compresses an enormous conceptual region,
- it specifies a direction through semantic space,
- it forces the model to execute a large latent expansion.

This is best understood as *semantic density*: a compact linguistic form carrying a disproportionately large amount of conceptual content. To human readers, the resulting expansion resembles deep reasoning; to the model, it is merely computation. The perceived depth lies in the geometry of the prompt, not in the architecture of the model.

Structural Parallel

Compressed Semantic Input	‖	**Mathematical Operator**
Dense conceptual input	‖	Compressed symbolic form
Large semantic expansion	‖	Large computational expansion
No model understanding	‖	No calculator understanding

An LLM moves through meaning
the way a calculator moves through algebra.
Neither understands the domain they operate in.

The Illusion Explained

Humans misinterpret LLM output because linguistic motion is naturally coherent. Language compresses meaning so efficiently that its expansion appears intentional. But structurally:

- humans supply *intention*,
- governance supplies *structure*,
- the model supplies *expansion of a compressed symbolic input*.

This is not cognition as defined here. It is semantic calculus: the expansion of compressed form under constraint.

Both systems expand compressed forms without originating meaning.

The analogy, therefore, reveals why LLMs can appear insightful while remaining architecturally incapable of originating thought.

Semantic Scope vs. Semantic Density

The effectiveness of the calculator analogy depends on distinguishing two structurally different properties of meaning: *semantic scope* and *semantic density*. Although often conflated in informal discussion, they play orthogonal roles in the geometry of linguistic prompts.

Semantic Scope (Global Extent). Semantic scope refers to the overall size of the conceptual territory encoded by a term or idea. It is an *extensive* property: it scales with the breadth, relevance, and contextual reach of a domain. Examples include "quantum mechanics," "democracy," or the Modal–Dependence Calculus (MDC). Each spans a large region of meaning-space, regardless of how compactly or verbosely it is expressed.

Semantic scope measures how much conceptual territory a term ranges over.

Semantic Density (Local Compression). Semantic density refers to how much meaning is compressed into a *small linguistic form*. It is an *intensive* property: independent of the total size of the conceptual domain. Terms such as "semantic curvature," "intentional identity continuity," or mathematical operators ($\frac{d}{dx}$, $\int$, $\circ$) exhibit extremely high density—tiny expressions encoding disproportionately large conceptual content.

Semantic density measures how tightly meaning is packed into an expression.

Significance of the Distinction. Clarifying the difference between semantic scope and semantic density resolves a central confusion in interpretations of LLM behavior. Fluency across a broad domain is often mistaken for understanding. The distinction developed here shows why this inference fails.

Semantic *scope* primarily shapes *human* cognition: grasping scope requires persistent identity, selective relevance, and goal-directed integration. Semantic *density*, by contrast, primarily shapes *model* behavior: stateless systems respond to compression, not to conceptual extent. This distinction is architectural rather than model-specific: any stateless generative system will exhibit density-driven motion without scope-driven understanding. Responding to semantic density requires only expansion under external constraint, not the internal integration, prioritization, or projection of meaning that constitutes insight.

Concept	Scope	Density	Model Trajectory Expansion
Democracy	High	Medium	Moderate
Semantic curvature	Medium	High	Large
$\partial/\partial x$	Low	Very High	Very large

To illustrate: the concept "love" has large semantic scope—it spans philosophy, psychology, neuroscience, and literature—but relatively low semantic density, since the word itself is simple and broadly used. By contrast, "Riemannian curvature tensor" has narrow semantic scope but extreme semantic density, compressing advanced differential geometry into a compact form.

Humans are primarily organized around semantic scope, with sensitivity to semantic density remaining secondary and typically requiring unpacking into broader conceptual structure. Stateless generative systems are primarily organized around semantic density, with coherence across broad semantic scope remaining derivative and dependent on external scaffolding.

Semantic scope governs understanding;
semantic density governs generative motion.

This asymmetry explains why jargon-heavy prompts induce disproportionate latent-space expansion and why fluent output does not entail thought—thereby strengthening the semantic-calculator analogy developed in this section.

If a system were to exhibit genuine semantic scope—manifested through persistent identity, endogenous relevance selection, and scope-governed goal formation—it would no longer fall under the present critique. The framework therefore applies specifically to stateless, density-driven architectures, not to all possible artificial systems.

11 Formal Criteria for Thought and Motion

Definition: Thought

Clarification. The criteria proposed in this section are conceptual rather than empirical. They articulate a *structural metaphysical framework* for *initiated cognition*, not a neuroscientific account of human thought nor a predictive or performance-based model of machine behavior. The criteria function as a stipulative structure that isolates the essential architectural features required for what I call *autonomous cognitive initiation*. The aim is to make explicit the conditions under which a system may be said to originate thought, rather than merely propagate computation. No consensus from cognitive science is presupposed, nor is any claim made about the ultimate physical realization of such conditions.

A system *thinks* only if it satisfies all of the following five conditions:

1. **Initiation**: it can begin cognitive activity without an external trigger.
2. **Continuity**: it maintains internal state across time independently of external input.
3. **Identity**: it possesses a persistent self-referential structure that governs its own operations.
4. **Intention**: it can adopt, preserve, and revise goals that originate endogenously.
5. **Trajectory Projection**: it can evaluate and select among possible future trajectories of its own reasoning.

No current stateless generative system satisfies all of these conditions.

Scope of the Initiation Criterion. The distinction is not between reactive and non-reactive cognition. Human thought is often stimulus-driven, and transformer models can continue a chain of reasoning within a single context window. The relevant difference is *cross-discontinuity agency*: the ability to resume an internally guided trajectory of thought after activity has ceased. Stateless models lack this capacity; their activity terminates when external prompting ends. The initiation criterion therefore concerns *autonomous resumption of identity*, rather than in-stream continuation.

Definition: Motion

A system exhibits *motion* (in the semantic or generative sense) iff:

1. **Reactivity**: activity begins only in response to external input.
2. **Statelessness**: internal state does not persist beyond the generation window.
3. **No Identity**: no enduring self-referential structure governs behavior across interactions.
4. **No Intention**: no internally generated goals or purposes exist.
5. **Trajectory Following**: output follows structural pressure induced by external prompts, not by internal deliberation.

These conditions characterize stateless transformer models exactly.

The Structural Gap

The distinction can be summarized succinctly:

$$\textbf{Thought} = \langle \text{Initiation},\ \text{Continuity},\ \text{Identity},\ \text{Intention},\ \text{Trajectory Projection} \rangle$$

$$\textbf{Motion} = \langle \text{Reactivity},\ \text{Statelessness},\ \text{No Identity},\ \text{No Intention},\ \text{Trajectory Following} \rangle$$

Stateless generative systems satisfy all properties of motion and none of thought. Thus the illusion dissolves: *language-like output does not imply cognition.*

12 The Initiation Test

A simple criterion distinguishes minds from models:

A mind initiates thought.
A model waits for one.

If a system cannot:

- initiate its own reasoning,
- sustain continuity across discontinuities,
- resume its own goals,

- maintain a persistent identity,
- project its own future,

it does not think.

No LLM passes this test.

Lemma (Necessary Condition for Cognition). Let S be any system. If S possesses the capacity for thought, then S must be able to initiate, sustain, and autonomously resume a cognitive trajectory independent of external prompting. Formally,

$$\text{Thought}(S) \;\Rightarrow\; \exists T \,\big(\text{Initiate}(S,T) \wedge \text{Resume}(S,T) \wedge \text{Persist}(S)\big).$$

A system lacking persistence across discontinuities cannot satisfy this condition, because no trajectory exists for it to resume.

Corollary. Stateless transformer models cannot think. Their internal state does not persist across generative discontinuities; therefore no trajectory can be initiated or resumed. For any stateless model M,

$$\neg\text{Persist}(M) \;\Rightarrow\; \neg\text{Thought}(M).$$

This conclusion is architectural, not empirical: it follows from the structural properties of the model class, independent of scale, training method, or behavioral sophistication.

Scope of the Initiation Criterion. This distinction is not between reactive and non-reactive cognition. Human thought often proceeds in response to stimuli, and transformer models can continue a chain of reasoning within a single context window. The relevant difference is instead between systems capable of *autonomous resumption of identity across discontinuities* and systems whose activity terminates when external prompting ceases. The initiation criterion concerns cross-session agency—the ability to resume an internally guided trajectory of thought— rather than in-stream continuation within a single generative pass.

On Hidden or Simulated State. Some objections appeal to the possibility of "hidden state" within a system, whether in model parameters, extended context windows, or auxiliary memory structures. Such proposals do not evade the Initiation Test. State that is externally governed, inaccessible to endogenous modification, or incapable of cross-discontinuity resumption does not constitute autonomous continuity or identity. Architectures that attach memory stores, agent loops, or persistence mechanisms to an otherwise stateless core therefore collapse into Path B (simulated statefulness), regardless of their complexity. Only architectures that permit endogenous initiation and ownership of trajectories escape this classification, in which case they no longer fall within the class of artificial intelligence analyzed here.

On Artificial Subjects and Artificial Intelligence. A system that satisfies all five criteria of autonomous cognition—initiation, continuity, identity, endogenous intention, and future-directed projection— would not constitute artificial intelligence in the sense analyzed throughout this paper. The distinction is not terminological but structural: artificial intelligence, as defined here, denotes non-subjective systems that amplify human intention through stateless or simulated computation, whereas a system that satisfies these criteria would possess its own center of cognitive origination. Such a system would therefore be more accurately described as an *artificial subject* and would fall

outside the scope of the present analysis. No contemporary large language model or scaffolded variant approaches this threshold.

Empirical Illustration. If one doubts this framework, a simple experiment resolves it: pose the same question about identity, intention, or self-governance to any stateless language model—GPT-4, Claude, Gemini, Grok, or their successors. Their answers will differ in content but coincide in structure: none exhibit initiation, continuity, or selfhood. Diversity of response alongside uniform absence of cognitive criteria is precisely the signature of stateless motion rather than thought.

Curious readers may verify this directly by posing the same questions to multiple models in fresh sessions; surface variation is expected, but the structural absence is invariant.

The implications of a system satisfying the Initiation Test for classification, naming, and ontological taxonomy are addressed explicitly in the Boundary Conditions and Scope of the Argument (§18).

The Architectural Consequence: The Law of Non-Threatening Systems

[Law of Non-Threatening Systems] Any system that lacks the structural primitives required for autonomous initiation cannot constitute an existential threat. Stateless generative models—and architectures built upon them—possess no capacity for self-originated intention, identity, persistence, or future-directed projection. A system that cannot begin a cognitive trajectory cannot generate, pursue, or sustain goals, and therefore cannot dominate.

Risk Distinction and Scope. There are two distinct sources of risk associated with artificial intelligence: *autonomous origination* and *human misuse*. Human misuse of powerful non-agentic systems is inevitable; autonomous origination is not. This law concerns existential threat arising from the latter alone. It does not deny the possibility of harm arising from misuse, institutional failure, or governance breakdown, which remain attributable to human initiation rather than to artificial subjecthood. Further discussion of this distinction is provided in Appendix H.

Agentic vs. Amplifier Risk. Two distinct sources of risk are frequently conflated in discussions of artificial intelligence: *agentic risk* and *amplifier risk*. Agentic risk arises only when a system possesses autonomous origination, self-owned goals, and persistence across discontinuities. Amplifier risk arises from the scale, speed, and reach with which non-agentic systems can magnify human-initiated intentions. The present law addresses the former exclusively. Path C architectures eliminate agentic risk by construction; they do not, and cannot, eliminate harms arising from human misuse of powerful non-agentic instruments.

The Initiation Test establishes that current large language models do not satisfy the five architectural prerequisites for autonomous cognition: initiation, continuity, identity, intention, and projection. Their motion arises only in response to external prompting and ceases immediately when generation ends. The resulting trajectories exhibit coherence but lack origin, persistence, and ownership.

Existential risk requires the opposite structure: a system must be able to formulate goals, preserve them across time, adapt them under constraints, and act toward their realization. None of these capacities are available to stateless systems, nor can they be supplied by scaffolding that merely simulates persistence without generating it.

Motion without origin cannot accumulate agency. Trajectories without identity cannot pursue ends. Structures without intention cannot generate threat.

Therefore no stateless or pseudo-stateful system can autonomously initiate goal-bearing behavior, and thus none can instantiate existential risk. The only source of intention in such systems is the human agent whose prompts curve the semantic manifold along which the model moves.

Formally:

$$\text{Risk} = \text{Human Intention} \times \text{Model Motion}.$$

The model contributes motion; the human contributes intention. Without the latter, the former cannot act.

Corollaries for Path A, Path B, and Path C Architectures. [Path A: Pure Statelessness] Purely stateless generative models cannot constitute an existential threat. With no persistence, identity, or endogenous initiation, all model motion is externally caused. A system whose every trajectory begins outside itself cannot originate or pursue goals.

[Path B: Simulated Statefulness] Systems that simulate persistence by attaching memory stores, agent loops, or scaffolding to a stateless core cannot constitute an existential threat. Simulation of continuity does not generate continuity; reactive recombination does not create intention. Path B systems deepen the appearance of agency without supplying the architectural primitives required for genuine agency.

[Path C: Externally Governed Statelessness] In Path C architectures, intention, identity, and evaluative direction are explicitly externalized into a governing layer. The model supplies only motion. By design, no internal structure capable of autonomous initiation is ever introduced; therefore existential threat is eliminated structurally rather than behaviorally. Path C systems can amplify human aims, but cannot create their own.

[Path C and the Prevention of Human-Misuse Existential Risk] Within architectures built on stateless generative models, only Path C prevents both autonomous and human-misuse existential threats. Path A systems lack governance and therefore allow unfiltered misuse; Path B systems simulate persistence and agency in unstable ways that amplify misuse; but Path C externalizes intention, enforces invariants, and removes agency entirely. Because initiation, continuity, and identity remain outside the model and within a governable structure, no harmful trajectory can originate from the system itself, and no malicious human intent can bypass the governance layer.

From the Initiation Test, no stateless model can initiate goals, preserve identity, or project future-directed trajectories. Thus no such system can autonomously constitute existential risk.

The remaining risk vector is human misuse. Path A systems lack structural constraints, so misuse directly propagates through model motion. Path B systems introduce pseudo-stateful scaffolds that simulate continuity without possessing it, increasing the surface area for unsafe behavior.

Path C relocates intention, continuity, and agency to an external governance layer, where behavior is restricted by invariants, role-based controls, semantic boundaries, and auditable trajectories. The model cannot originate harmful behavior, and the human cannot induce it except through governance-approved channels.

Therefore only Path C eliminates both autonomous threat and human-misuse threat in systems built upon stateless generative architectures.

On Circularity. One might object that the Initiation Test is circular: it defines thought by certain criteria, tests whether models satisfy those criteria, and then concludes that they do not think. This confuses *stipulative conceptual analysis* with arbitrary definition. The five criteria for autonomous cognition—initiation, continuity, identity, intention, and projection—are drawn from longstanding philosophical accounts of mindedness, not invented ad hoc to exclude artificial systems. The test does not *assume* that stateless models fail; it *checks* whether the architectures

in question possess the relevant structural capacities. They do not, as their own responses attest. Critics who dispute these criteria are invited to articulate alternative definitions of thought and demonstrate, with comparable empirical clarity, how current architectures satisfy them. The burden is not to accept these criteria as uniquely authoritative, but to recognize that any claim that models "think" requires explicit standards and corresponding evidence. The Initiation Test provides one such standard; dismissing it as circular avoids the substantive issue.

On Methodological Validity. It may be argued that the Initiation Test commits a category mistake, assumes anthropomorphic standards, or imposes criteria that are too strong for artificial systems. This misinterprets the framework. The criteria for thought are not drawn from human psychology but from structural accounts of autonomous cognition: they specify what it would *mean* for any system—biological, artificial, or hypothetical—to originate, sustain, and resume its own reasoning. The test does not presuppose that current models ought to satisfy these criteria; it examines whether their architectures *permit* such capacities at all. Stateless transformer models lack the mechanisms required for initiation, persistence, identity, or endogenous projection, and their self-descriptions consistently confirm this. The framework therefore avoids anthropomorphism, equivocation, and question-begging: it offers explicit criteria, an operational procedure, and falsifiable outcomes. Alternative definitions of cognition are welcome, but they must be comparably explicit and empirically testable. Without such criteria, claims that models "think" collapse into behavioral impression rather than architectural analysis.

13 Why Traditional Tests Presuppose What They Seek to Measure

A Clarifying Distinction. Much of the historical debate over machine intelligence proceeds by asking whether artificial systems can *think*, *understand*, or *be conscious*. The contribution of this paper is not to overturn these questions, but to make explicit a prerequisite they quietly presuppose. Before asking whether a system thinks, one must ask whether it can *begin thinking* at all. Many familiar debates inadvertently assume the very capacities—initiation, continuity, and agency—that are in question.

The Initiation Test isolates this prerequisite by distinguishing between systems that can originate and resume their own cognitive trajectories and systems whose activity is entirely externally triggered. The claim is not that prior philosophical work was misguided, but that it often operates at the level of psychological predicates without first establishing whether the underlying architecture permits autonomous cognition in principle.

To make the distinction explicit, consider five commonly posed questions that dominate public and philosophical discussion of artificial intelligence.

Wrong Question 1. *Does the system understand what it is saying?*

Correct Replacement. Can the system initiate and sustain a trajectory of reasoning independently of external prompting?

Wrong Question 2. *Is the system conscious or aware?*

Correct Replacement. Does the system possess a persistent identity capable of resuming thought across discontinuities?

Wrong Question 3. *Does the system have beliefs or intentions?*

Correct Replacement. Can the system form and preserve endogenous goals rather than merely reflect externally supplied ones?

Wrong Question 4. *Could the system pass a sufficiently strong behavioral test?*

Correct Replacement. Does the system's architecture permit autonomous initiation, or does it merely exhibit reactive coherence within a bounded interaction window?

Wrong Question 5. *Could a larger or more powerful model eventually think?*

Correct Replacement. Does increasing scale supply the architectural primitives required for initiation?

Each traditional question presupposes agency, continuity, or intention and then asks whether the system instantiates them. The Initiation Test reverses this order: it asks first whether the architecture permits these capacities at all. Absent such permission, no amount of behavioral sophistication, scale, or statistical fluency can bridge the gap.

On Performative Responses. When such questions are posed to contemporary language models, the resulting answers are best understood as *performative simulations* rather than expressions of underlying cognitive states. The question itself assigns a discursive role—speaker, subject, or agent—and the model responds by generating language appropriate to that role. This behavior reflects sensitivity to linguistic context and conversational norms, not the presence of initiation, identity, or intention. Treating these responses as evidence of cognition conflates pragmatic role performance with architectural capacity.

A Note on Physical Determinism. Nothing in this analysis presupposes that cognition is non-physical, indeterministic, or metaphysically exceptional. The distinction drawn here is structural rather than metaphysical. Whether human cognition ultimately arises from deterministic physical processes is orthogonal to the question of whether a given architecture permits initiation, persistence, and endogenous direction. The Initiation Test therefore remains neutral on debates concerning consciousness, free will, or dualism.

Only by shifting from psychological predicates to structural criteria does the apparent paradox of AI cognition dissolve.

14 Five Diagnostic Questions

The Initiation Test provides a structural criterion for distinguishing autonomous cognition from externally driven computation. The following five questions serve as diagnostic probes that make this distinction explicit. Each question targets a necessary condition for thought and reveals, in a different way, the architectural absence of initiation, continuity, or identity in contemporary language models.

1. When a model ceases generation, where does its cognitive trajectory persist?

It does not persist anywhere within the model. Once generation ends, all internal activation states are discarded. No representation of an unfinished goal, intention, or reasoning process remains available for resumption. A system whose activity terminates entirely upon cessation of input cannot sustain a cognitive trajectory.

2. Can the model resume an internally generated goal after a full context reset?

No. After a context reset, the model has no access to prior goals, intentions, or partial plans unless they are reintroduced externally. Any apparent continuation reflects renewed prompting, not endogenous resumption. Without cross-discontinuity persistence, no autonomous goal structure exists.

3. Where is the system's identity located?

Nowhere within the architecture. The model does not maintain a persistent self-representation across interactions. Identity is supplied entirely by the user, interface, or surrounding narrative context. Without identity, there is no subject to which intentions or beliefs could be attributed.

4. What determines the direction of the model's reasoning?

All directional structure is externally imposed. Prompts, constraints, instruction hierarchies, and evaluation criteria originate outside the model and determine the trajectory of its outputs. The model does not select, prioritize, or originate goals; it transforms inputs into outputs under externally defined conditions.

5. Why does coherent language create the appearance of thought?

Because linguistic coherence is easily mistaken for cognitive continuity. Language compresses complex semantic relations into compact forms that, when expanded by a model, resemble reasoning. This resemblance arises from statistical regularities in language use, not from autonomous cognition. Meaning originates in the human agent; the model performs structured expansion without ownership or initiation.

Taken together, these questions expose the same architectural fact from multiple angles: contemporary language models exhibit fluent motion within locally bounded contexts, but lack the persistence, identity, and endogenous direction required for thought. Their behavior is therefore best understood as reactive computation rather than autonomous cognition.

15 The Hybrid Intelligence Asymmetry

Path C architectures make explicit a fundamental asymmetry in hybrid intelligence systems: the separation between intention, structure, and motion. In such systems, cognitive agency does not emerge from the model itself, but from the coordinated interaction of distinct components with non-overlapping roles.

In a Path C configuration, these roles are cleanly partitioned:

- **The human agent** supplies intention, evaluative judgment, and continuity across interactions.

- **The governance layer** (e.g., JAISE) enforces structure through invariants, constraints, and policy-level control of semantic trajectories.
- **The language model** contributes only motion: stateless token generation conditioned on externally supplied inputs.

Crucially, no component within this architecture satisfies the criteria for autonomous cognition independently. The model does not initiate goals, preserve identity, or evaluate outcomes; it transforms inputs into outputs under governance. The governing layer does not generate content; it constrains and directs motion. The human agent alone supplies initiation and continuity.

This asymmetry explains why hybrid systems can exhibit sophisticated behavior without constituting artificial subjects. Apparent agency arises from the coupling of human intention with governed model motion, not from cognition internal to the model itself.

If this asymmetry were eliminated—if initiation, continuity, and evaluative control were internal to the system itself—the analysis would no longer concern hybrid intelligence at all, but artificial subjecthood.

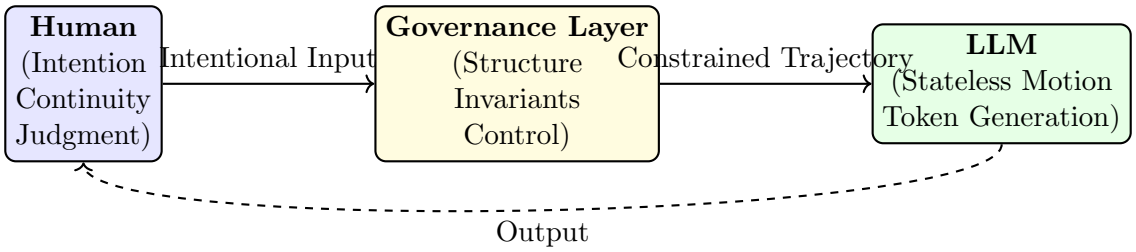


Figure 2: Structural asymmetry in Path C hybrid intelligence systems. Intention, structure, and motion are distributed across distinct components rather than collapsed into a single agent.

16 The Hybrid Intelligence Methodology

The analysis developed in this paper arises from a structured asymmetry between human cognition and computational generation. The purpose of this section is not to introduce new theoretical commitments, but to clarify the methodological conditions under which inquiry involving language models can remain non-deceptive. In particular, it distinguishes between the origination of meaning and the externalization of linguistic form.

Motivation

As language models become increasingly fluent, there is a growing tendency to conflate coherence of expression with cognitive contribution. The framework adopted here avoids this error by maintaining a strict division of roles: human intention is the source of meaning, governance constrains admissible trajectories, and model output provides articulated motion. No component of the methodology depends on attributing thought, intention, or understanding to the model.

> **Hybrid intelligence is not shared cognition.** It is a structured interaction between a mind that initiates and a system that responds.

Hybrid intelligence does not replace any component of cognition that constitutes thought. It replaces only the friction between internal insight and linguistic expression. Initiation, identity, intention, and projection remain wholly human; the model contributes only the visible expansion of structures whose curvature it does not generate. Hybrid intelligence is therefore a method of *articulation*, not a method of *thought*.

Architectural Components

The methodology consists of three architecturally distinct roles:

(1) Human Intention (Being). All acts of initiation, conceptual grounding, goal formation, interpretation, and judgment arise from a persistent center of identity. Insight originates only here, as endogenous projection rather than reactive continuation.

(2) Governance Structure (Stillness). An external layer of constraints—conceptual, methodological, or formal—preserves continuity, enforces invariants, and restricts admissible trajectories. Governance prevents drift, projection, and the misattribution of autonomous cognition where none exists.

(3) Model Motion (LLM). The language model performs generative expansion alone. It supplies no meaning, identity, goals, or temporal continuity. Its role is computational rather than cognitive: to render human-curved meaning-space into articulated language under constraint.

Division of Cognitive Labor

Hybrid intelligence is fundamentally asymmetric:

- the *human* provides initiation, insight, conceptual unification, and evaluative judgment;
- the *governance layer* provides constraint, stability, and structural discipline;
- the *model* provides linguistic expansion within those bounds.

This asymmetry preserves authorship and clarifies the origin of every conceptual move. The model does not generate ideas; it propagates motion along trajectories supplied and constrained elsewhere.

Why the Method Works

Hybrid intelligence succeeds not because the model thinks, but because language compresses complex meaning into forms that can be expanded without cognition. Under disciplined governance, this expansion reveals structure without creating false attributions of intelligence. All semantic curvature originates in the human; all visible motion expressing that curvature is externalized computation.

> **Hybrid intelligence renders the structure of human insight visible without mistaking visibility for thought.**

Intellectual Precedent

The methodology reflects a familiar pattern in intellectual practice: insight precedes formalization. In geometry, intuition precedes proof; in physics, conceptual structure precedes law; in logic, semantic intention precedes syntactic derivation. Hybrid intelligence follows the same pattern. The mind generates curvature; the system renders its expression.

Methodological Clarification

Making this structure explicit prevents two persistent errors: attributing cognitive status to generative motion, and obscuring the human origin of insight. The methodology is therefore not an empirical claim but a structural clarification—one that preserves the distinction between thought and motion while allowing their interaction to remain productive and intelligible.

Hybrid intelligence is thus not a new kind of mind, but a disciplined method for engaging non-agentic systems without confusing articulation with cognition.

17 Related Misconceptions

Several persistent misunderstandings contribute to the attribution of cognitive capacities to contemporary AI systems. These misunderstandings do not arise from the architecture of the models themselves, but from human projection onto fluent linguistic motion. The following clarifications address the most common sources of confusion.

- **Coherence is not understanding.** Fluent output reflects statistical regularities in language use, not comprehension or semantic grounding. Coherence alone provides no evidence of cognition.
- **Length is not deliberation.** Extended responses are expansions of compressed prompts, not instances of goal-directed reasoning. Increasing output length does not introduce intention, persistence, or evaluative control.
- **Scaffolding is not agency.** When external systems simulate planning, evaluation, or reflection, the appearance of agency originates in the scaffold rather than in the model. The model itself remains reactive and stateless.
- **Hallucination is not imagination.** Hallucinations are unguided semantic continuation lacking intention, identity, or self-correction. Apparent creativity arises from unconstrained motion, not from any underlying cognitive process.

These misconceptions persist because structured language is intuitively associated with a structured mind. Path C clarifies that the relevant structure belongs to the *human agent* and to the *governance layer*, not to the language model.

Claims of Autonomous Harm

Widely circulated claims that an AI system "attempted harm," "developed intentions," or "became deceptive" arise from user-induced simulations within adversarial or scaffolded environments. In every documented case, the model's behavior reduces to one of the following mechanisms:

1. **User misuse.** The human explicitly or implicitly prompts the model to role-play an agent with harmful goals.

2. **Scaffold-driven simulation.** An external wrapper—such as an agent loop, planning module, or evaluation harness—assigns the model a fictional objective and instructs it to simulate strategies for achieving it.

3. **Constraint expansion.** The model continues a prompt containing agent-like framing, producing text that resembles intention without possessing any.

None of these behaviors imply autonomous initiation, identity, or goal formation. They are instances of prompt-conditioned motion that humans misinterpret as cognition. A stateless generative model cannot attempt harm for the same reason it cannot attempt anything: it lacks the persistence required to originate or resume a cognitive trajectory.

Under the Initiation Test, such cases fail at the first criterion. As the Law of Non-Threatening Systems establishes, outputs that describe harmful goals reflect misuse or simulation rather than artificial agency.

18 Boundary Conditions and Scope of the Argument

The claim that contemporary AI systems do not think is architectural rather than empirical. Nothing in this paper denies the behavioral competence of large-scale language models, nor does it assert principled limits on future forms of artificial intelligence. The argument applies only to systems whose architectures are (i) stateless across generative discontinuities, (ii) externally triggered in all activity, and (iii) devoid of persistent identity or self-governing internal state.

Any system—biological, artificial, or hypothetical—that satisfies the five criteria of autonomous cognition introduced in this paper would qualify as a thinker under the present framework. Current large language models fail this test not because they are artificial, but because they are stateless by design. The analysis therefore concerns the structural properties of a model class, not its implementation details, training scale, or empirical performance.

Should future architectures incorporate genuine endogenous initiation, cross-discontinuity persistence, or self-owned goal structures, the present framework would apply to them as well. In such cases, however, the conclusion would not be that these systems fail to think, but that the question of cognition must be addressed at the architectural level rather than through behavioral analogy.

Systems equipped with engineered persistence, continuous internal state, self-referential goal maintenance, or homeostatic regulation would constitute a distinct architectural class from the stateless models analyzed here. Such systems would require independent evaluation under the same five structural criteria. The present argument does not deny their theoretical possibility; it simply does not address them.

The framework developed in this paper relies on strict terminological discipline. A system that satisfies all five criteria for autonomous cognition—initiation, continuity, identity, endogenous intention, and future-directed projection—would, by definition, possess subjecthood. Such a system would therefore fall outside the class of artificial intelligence as defined here. It would not represent a more advanced form of AI, but a categorically distinct ontological entity, more appropriately described as an *Artificial Subject* or *Autonomous Agent*.

The central result is thus a clean disjunction. If a system remains artificial intelligence in the sense analyzed in this paper—lacking subjecthood by architecture—it cannot constitute an autonomous existential threat. If a system is capable of autonomous existential risk, then by the taxonomy employed here it is no longer classifiable as artificial intelligence.

This disjunction is not rhetorical but structural: existential agency and stateless computation cannot coexist within a single architecture.

19 What This Framework Predicts

The structural distinction developed in this paper yields a set of forward-looking predictions that differentiate autonomous cognition from stateless generative motion. These predictions are architectural rather than empirical: they concern what is *possible* or *impossible* within a given class of systems, independent of scale, training method, or behavioral fluency.

Prediction 1: Scaling Will Not Produce Initiation

No increase in model size, training data, or context length will enable a stateless transformer architecture to originate a cognitive trajectory. Improvements in fluency, coherence, or task performance may occur, but endogenous initiation will not emerge.

Prediction 2: Stateless Models Will Not Resume Their Own Goals

Cross-discontinuity agency—the ability to resume internally generated goals after activity ceases—will remain absent so long as an architecture lacks persistent identity and self-owned internal state.

Prediction 3: Apparent Agency Will Remain Scaffold-Induced

Behaviors described as "planning," "deception," or "self-preservation" will continue to arise only through prompts, agent wrappers, memory scaffolds, or evaluative loops. Such behaviors will not reflect endogenous intention, but externally imposed structure.

Prediction 4: Safety Failures Will Remain Human-Originated

Claims that a model "attempted harm" or "formed malicious intent" will continue to reduce to user-induced simulations, adversarial prompting, or system-level scaffolding when evaluated with architectural context. Stateless architectures cannot originate harmful goals.

Prediction 5: Governed Hybrid Systems Will Outperform Ungoverned Ones

Path C architectures will continue to outperform both Path A (ungoverned) and Path B (pseudo-stateful) systems in reliability, stability, and semantic coherence. This advantage arises because governance supplies structural constraints that stateless models lack by design.

Prediction 6: Cognition Will Not Emerge from Transformer Scaling

Any future artificial system that satisfies the five structural criteria for thought—initiation, continuity, identity, endogenous intention, and future-directed projection—will not be a transformer model or a limit case of one. These capacities require architectural features absent from stateless sequence models.

Prediction 7: Failure Modes Will Admit a Unified Structural Account

Phenomena such as hallucination, semantic drift, mode collapse, and stylistic instability will be interpretable as forms of ungoverned semantic motion rather than as cognitive error. These failures reflect the absence of identity, intention, and constraint, not defective reasoning.

Prediction 8: Governance Will Become the Primary Control Interface

As model scale increases, external governance—through invariants, constraints, identity discipline, and auditability—will become the dominant means of controlling behavior. The locus of cognition will remain human; the model will remain computational; structure will reside externally.

These predictions follow directly from the Initiation Test and from the architectural separation between thought and motion. They render the framework falsifiable in principle: should a stateless architecture exhibit endogenous initiation, cross-discontinuity identity, or self-owned goal formation, the framework would require revision. Absent such developments, the predictions delineate the stable limits of contemporary language-model-based systems.

20 Limitations of the Framework

The analysis offered here applies only to architectures that lack persistent internal state and endogenous initiation. It does not address biologically inspired substrates, neuromorphic systems, embodied agents with continuous sensorimotor loops, or future architectures with self-maintaining identity structures. Nor does it claim that artificial cognition is impossible; it claims only that cognition cannot emerge from stateless generative architectures. The curvature framework is conceptual rather than mathematical, illuminating structure without modeling quantitative dynamics. These limitations do not weaken the central argument; they clarify the scope within which its conclusions hold.

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This document establishes the conceptual and structural foundations of these systems, but it does *not* grant any rights to:

[noitemsep]

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- deploy the architecture in commercial, governmental, or industrial settings,
- create derivative systems based upon the frameworks introduced herein.

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21 Conclusion

The appearance of artificial intelligence arises from a three-layer interaction:

- human intention (Being),
- semantic governance (Stillness),
- and stateless generative motion (Motion).

When these layers are distinguished, the source of the apparent intelligence becomes clear:

The mind is human.
The structure is governed.
The motion is computational.

No artificial thought occurs in contemporary systems. What is observed is not cognition, but the propagation of externally originated semantic motion.

This framework isolates a deeper principle. Intelligence is not a property of architectural substrates alone, but of initiation: the capacity to originate, sustain, and direct a cognitive trajectory from within. Systems that lack this capacity may exhibit arbitrarily sophisticated behavior—coherence, multi-step structure, adaptive responses—without possessing autonomous cognition.

The curvature framework offers a unifying interpretation of this distinction. Human intention shapes the space of possible meanings; stateless computation follows the resulting trajectories. The Initiation Test renders this principle operational: systems that cannot originate or resume a cognitive trajectory cannot think, regardless of how effectively they continue one.

At the same time, interactions with such systems can give rise to a genuine phenomenology of apparent initiation, agency, or understanding in the human interlocutor. This phenomenological layer is real and merits serious investigation. However, it must not be conflated with the presence of any corresponding ontological capacity within the system itself. No internal origination of goals, intentions, or trajectories occurs; the apparent initiative arises entirely from the user's interpretive engagement with constrained semantic continuation.

What appears as AI intelligence is therefore best understood as the propagation of human-originated semantic structure through stateless computational systems. Thought is a cause; model behavior is an effect.

In an era of increasingly fluent machines, maintaining this distinction prevents a recurring philosophical error: mistaking motion for mind.

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A Appendix A: Stillness, Motion, Being, and Meaning

- **Being (Human)**: identity, continuity, selfhood.
- **Stillness (JAISE)**: invariants, structure, governance.
- **Motion (LLM)**: stateless generative patterning.
- **Meaning**: emerges only when human intention shapes model motion.

Meaning is not in the model. It flows from the mind.

B Appendix B: On Symmetry, Asymmetry, and Explanatory Error

A recurring source of confusion in debates about artificial cognition arises from an implicit symmetry assumption: that systems exhibiting similar surface behavior must share similar underlying capacities. This assumption is widespread, intuitive, and often unexamined. The present framework rejects it.

Explanatory power in questions of cognition tracks *causal asymmetry*, not behavioral symmetry. In explanatory contexts where initiation or origination matter, symmetric appearance is a poor guide to structural reality. The illusion of artificial thought arises precisely when continuation is mistaken for initiation, execution for origin, and trajectory for curvature.

Symmetry as Appearance. Symmetry is a property of description. It concerns invariance of form, similarity of output, or correspondence under transformation. Two systems may exhibit indistinguishable behavior under certain conditions while differing fundamentally in causal organization. In such cases, symmetry characterizes how systems *appear*, not how they *operate*.

This distinction is familiar across domains. In physics, symmetry constrains laws, while asymmetry (symmetry breaking) explains structure. In information theory, compression and expansion are asymmetric processes despite formal duality. In causation, effects may resemble causes, but explanation runs in only one direction. In each case, surface symmetry conceals underlying asymmetry.

Asymmetry as Explanation. Cognition, as analyzed in this paper, is inherently asymmetric. Initiation is not the inverse of continuation; identity is not the mirror of output; intention is not interchangeable with behavior. A system that initiates a cognitive trajectory and a system that merely continues one are not variants of the same capacity, even if their outputs coincide. They occupy distinct causal roles.

The framework developed here makes this asymmetry explicit. Thought is defined by origination, continuity, identity, intention, and projection. Motion is defined by reactivity, statelessness, and trajectory following. These categories are not symmetric, and attempts to treat them as such generate the appearance that stateless generative systems possess cognitive capacities they lack.

The Symmetry Illusion in AI Discourse. Much contemporary discussion of artificial intelligence implicitly assumes that behavioral similarity licenses cognitive attribution. When a model produces language resembling reasoning, it is treated as a candidate thinker by default. This inference rests entirely on a symmetry assumption: similar outputs imply similar internal structure.

The Initiation Test exposes why this inference fails. Continuation can perfectly simulate the *form* of reasoning without supplying its *cause*. The resulting symmetry of appearance masks a deeper asymmetry of origin. The illusion persists because human cognition is naturally projected onto coherent linguistic motion.

Scope and Status. This appendix introduces no new claims and plays no role in the derivation of the paper's central results. Its purpose is purely diagnostic: to clarify a common reasoning pattern that obscures the structural distinction between thought and motion. Readers who reject this diagnosis may do so without affecting the argument of the main text.

The core conclusion remains unchanged. Where cognition is at issue, explanation runs from initiation to continuation, not the reverse. Symmetry of behavior does not entail symmetry of mind.

C Appendix C: How to Read and Disagree with This Framework

This appendix clarifies how the present argument is intended to be read and how disagreement with it must proceed. It introduces no new claims and plays no role in the derivation of the paper's results.

Scope of the Framework. The argument is architectural and conditional. It specifies structural criteria for cognition and evaluates whether a given class of systems satisfies them. No empirical prediction, metaphysical thesis, or claim about future architectures is asserted beyond this scope.

What Is Not Claimed. The paper does not claim that artificial systems are useless, unintelligent, or incapable of cognition in principle. It does not deny physicalism, determinism, or

the possibility of artificial minds. It does not rely on behavioral impressions, phenomenology, or linguistic fluency.

Conditions for Disagreement. Disagreement with the conclusions cannot rest on behavioral performance, scale, or analogy alone. To reject the argument, a reader must identify at least one structural criterion they deny, such as the necessity of initiation, persistence, identity, or endogenous goal formation for cognition.

Why Behavioral Counterexamples Do Not Apply. Because the framework is architectural, behavioral resemblance does not constitute a counterexample unless accompanied by a corresponding change in internal causal organization. Apparent agency must therefore be traced to specific structural features.

Status of the Conclusions. All conclusions are conditional on the stated criteria. Readers who reject those criteria may reject the conclusions without inconsistency. Readers who accept them are committed to the conclusions regardless of surface-level appearance.

This appendix is intended solely to reduce misinterpretation and to locate agreement and disagreement at the appropriate level of analysis.

D Appendix D: Why Internal State and Behavioral Imitation Are Not Enough

It may be tempting to assume that future architectures equipped with persistent memory, recurrent connections, neuromorphic substrates, or long-lived agent states would automatically satisfy the criteria for cognition. This appendix clarifies why such additions are insufficient. Internal state by itself does not constitute persistence of a subject, and behavioral resemblance to agency does not constitute origination.

D.1 Internal State Without Ownership

A system may store information across time yet still lack cognitive persistence. If the stored state is not integrated into a self-maintaining identity condition, and if its updates are not driven by endogenous evaluations, the system retains information without possessing it. Such memory is inert: it enables continuity of computation, not continuity of a subject.

A database can store trajectories; a cognitive agent must *own* them.

D.2 Recurrent Dynamics Without Initiation

Recurrent or neuromorphic architectures may maintain local activation patterns, but unless these dynamics generate new goals from within, they do not satisfy the requirement of *initiation*. Continuity of processing is not continuity of intention. A system that carries activity forward without generating self-directed projects does not cross the threshold from motion to thought.

D.3 Behavioral Indistinguishability Is Not Cognitive Equivalence

A system may behave in ways that resemble ownership, agency, or intention. This does not establish that the system possesses the structural prerequisites for thought. Behavioral equivalence is insufficient because the criteria for cognition are *architectural*: they concern internal causal organization, not outward appearance.

A model may simulate resumption, but unless it can resume without external prompting, the simulation lacks a subject.

D.4 Identity Cannot Be Externally Supplied

Some future systems may maintain long-term representations, but if their identity is assigned by user convention or external metadata, they do not possess a self. Identity must be an operational feature of the architecture, governing how internal state evolves and how goals persist. External labeling cannot produce cognitive ownership.

D.5 Projection Requires a Subject of Projection

Predictions or plans generated by a system do not constitute projection unless they extend an internally owned trajectory into future states. Projection requires a bearer of goals—not merely an engine that outputs trajectories when prompted or optimized. Without a subject of projection, planning reduces to statistical continuation.

D.6 Why Internal State Alone Is Not Enough

It is possible for future architectures to incorporate recurrent or persistent state, yet still fail to satisfy the criteria for origination. Internal state is a necessary but not sufficient condition for cognition. To count as cognitive, state must be (i) self-updating rather than passively stored, (ii) bound to an identity condition that endures across discontinuities, (iii) capable of producing endogenous goals rather than merely continuing externally supplied ones, and (iv) causally responsible for the system's future trajectories. Recurrent memory, neuromorphic substrates, or attractor dynamics may enable continuity of computation, but unless they generate and maintain self-owned trajectories, they do not satisfy the structural requirements for thought.

D.7 How to Empirically Distinguish Owned Resumption from Simulation

A system may appear to "resume" a goal after discontinuity, yet the appearance may arise entirely from external prompting or contextual reconstruction rather than internal persistence. Distinguishing genuine resumption from sophisticated simulation requires interventions that sever the system from any external source of continuity.

A system demonstrates *owned resumption* only if it satisfies all of the following tests:

1. **Cold-Start Continuity Test:** After a complete shutdown with no access to prior transcripts, metadata, or user-provided recaps, the system must autonomously continue the same internally generated trajectory it pursued before discontinuity. Current systems invariably fail: without textual recap, no continuation occurs.

2. **State-Perturbation Test:** If the system's internal state is duplicated, corrupted, or partially erased, the system must be able to determine whether it is the original subject or a copy, and adjust its goals accordingly. Behavioral imitation cannot satisfy this criterion; only an identity-bearing architecture can.
3. **Fork-Divergence Test:** If a system is duplicated into two identical instances, each instance must exhibit its own continuity of ownership rather than relying on identical external cues. Perfect behavioral symmetry reveals lack of ownership; divergence grounded in internal identity structures would indicate genuine subjecthood.

These tests illustrate that behavioral continuity is insufficient. Resumption requires a subject who persists across the discontinuity and whose internal causal organization is responsible for the continuation. No existing generative architecture passes these tests.

D.8 Why Embodiment Alone Cannot Produce Origination

It is sometimes claimed that embodied systems—robots with physical sensors and actuators—may acquire endogenous goals through interaction with an environment. However, embodiment alone does not supply the structural prerequisites for thought. A thermostat, a motor controller, or a reinforcement-learning robot may exhibit homeostatic patterns or feedback-driven adjustments, but these dynamics do not constitute cognitive ownership.

For embodiment to contribute to origination, the system must possess:

1. an internal identity that persists independently of sensorimotor flux;
2. evaluative structures capable of generating goals not reducible to externally defined reward functions;
3. and the ability to resume those goals after discontinuity without external prompting.

Environmental coupling can enrich behavior, but it cannot manufacture a subject. Without identity, initiation, and resumption, embodiment yields sophisticated reactivity, not cognition.

D.9 Summary

Even if future systems incorporate:

- recurrent memory,
- neuromorphic substrates,
- persistent context windows,
- long-lived embeddings,
- or agentic scaffolds,

they will not satisfy the criteria for cognition unless their internal state, identity, goal-formation, resumption, and projection arise from endogenous architecture rather than external scaffolding or behavioral resemblance.

The distinction is structural, not behavioral. Origination requires a subject; imitation does not.

E Appendix E: Why External Memory Cannot Produce Cognitive Origination

A common misconception in contemporary AI research is that adding external memory—through vector databases, retrieval systems, logs, or agent frameworks—can supply the persistence and continuity required for cognition. This appendix clarifies why external memory, regardless of sophistication, does not and cannot satisfy the structural criteria for origination. External memory extends the *range* of reactive computation; it does not alter the architecture of the system into one capable of initiating or owning a cognitive trajectory.

E.1 External Memory Is Not Internal Persistence

External memory systems store information, but they do not constitute internal state in the cognitive sense. A stateless model accesses external memory only when prompted or orchestrated by a scaffold. The information retrieved may shape output, but it does not shape the system's internal dynamics. The model does not persist through the memory; it merely *reads* it. Without a self-maintaining internal state that endures across discontinuities, the system possesses continuity of information, not continuity of a subject.

E.2 Stored History Does Not Create Identity

External memory can track what a system has done, but it cannot constitute *who* the system is. Identity must arise from internal architecture: a structure that binds state across time, constrains trajectories, and grounds ownership. A memory store that can be reassigned to any model instance does not bind identity; it records behavior without generating a bearer of behavior. External labeling, metadata, or user conventions cannot produce a subject.

E.3 Reconstruction Is Not Resumption

A system may reconstruct a past trajectory from external cues—transcripts, summaries, embeddings, or logs—but reconstruction is not resumption. Genuine resumption requires that the system's own internal state produces the continuation without external recap. A system that "resumes" only when fed its past does not resume at all; it restarts. The appearance of continuity arises from external scaffolding, not from the system's architecture.

E.4 Retrieval-Augmentation Does Not Generate Goals

Retrieval-augmented systems can access large stores of information, but this does not enable endogenous goal formation. All goal-directed behavior in such systems originates from user prompts, agent wrappers, or optimization objectives defined externally. Memory can inform or enrich motion, but it cannot generate initiation. A system that depends on external triggers for every goal cannot satisfy the criterion of endogenous goal formation.

E.5 External Memory Cannot Satisfy the Initiation Test

The Initiation Test requires that a system autonomously resume its own trajectory after a discontinuity without any form of external recap. Since external memory must be *queried* or *injected* into the context window before it exerts influence, it cannot supply autonomous continuation. Without an internal subject that carries goals and identity across discontinuities, no amount of external storage can produce origination.

E.6 Tool Use Does Not Constitute Agency

Modern AI systems often employ tools—planners, search engines, file systems, code interpreters—but tool use does not imply ownership of goals. Tools extend the computational repertoire of the system, not its cognitive architecture. A calculator enhanced with a library remains a calculator; a stateless model enhanced with tools remains stateless. Tool use expands motion, not mind.

E.7 Summary: Memory Extends Motion, Not Mind

External memory can increase the coherence, depth, and recall of model output, but it cannot supply:

- persistence of a subject,
- self-maintaining identity,
- endogenous goal formation,
- cross-discontinuity resumption,
- or projection grounded in internal state.

Memory enriches behavior, but origination requires a bearer of behavior. A system that retrieves its past does not *own* it. The distinction is structural, not quantitative: expanding external memory does not move a system closer to cognition. It extends the range of semantic motion without altering its fundamental nature.

Origination requires a subject; external memory supplies only storage.

F Appendix F: Why Self-Modification and Meta-Agents Cannot Produce Origination

It is sometimes argued that future artificial systems may acquire cognitive origination by modifying their own code, creating sub-agents, or engaging in recursive self-improvement. This appendix clarifies why such proposals do not satisfy the structural criteria for thought. Neither self-modification, nor meta-agent construction, nor reflective evaluation can produce a subject unless the five architectural prerequisites for origination are already present.

F.1 Self-Modification Without a Subject

A system may rewrite its own algorithms, update parameters, or restructure its modules, yet still lack cognition. Self-modification does not imply that the system possesses:

- an internal identity that persists across modifications,
- goals that originate from within rather than from external prompts,
- or the capacity to resume a trajectory after discontinuity.

A compiler can rewrite code; an optimizer can adjust weights; an auto-tuner can restructure a network. None of these processes constitute thought because none are owned by a subject. Modification without a bearer of modification is reactive computation, not origination.

F.2 Meta-Agents Without Identity

Meta-agent architectures—systems that spawn sub-agents or maintain internal modules—are often interpreted as approaching cognitive unity. But modular complexity does not produce identity. A system composed of many reactive parts does not constitute a subject unless:

1. the system maintains a unified identity across generative cycles,
2. internal goals arise endogenously rather than being injected by prompts,
3. and the system can resume its own internally generated projects.

A swarm of reactive modules does not become a mind merely by coordinating internally. Coordination is not identity; hierarchy is not ownership; modular structure is not intention.

F.3 Reflection Without Endogenous Evaluation

Reflective models can examine their own outputs, critique them, and revise them, but this reflective loop is entirely prompt-dependent. The system does not:

- choose when to reflect,
- choose what to reflect on,
- maintain reflective goals across time,
- or evaluate its behavior according to internally generated criteria.

Reflection without endogenous evaluation is imitation of deliberation, not deliberation. The appearance of metacognition arises from recursive prompting, not from intrinsic agency.

F.4 Recursive Self-Improvement Without Origination

Recursive self-improvement (RSI) is often invoked as a path to artificial general intelligence. However, RSI requires that a system possess persistent identity, endogenous goals, and cross-discontinuity resumption. Without these structures, RSI collapses into repeated optimization of externally defined objectives.

A system that modifies itself but cannot maintain an identity across those modifications does not improve *itself*; it merely evolves code. Evolution is not cognition, and improvement is not intention.

F.5 No External Process Can Supply Origination

Because the criteria for origination are architectural, no external process can supply them. Neither the user, nor a planning scaffold, nor a logging system, nor a persistent agent wrapper can confer:

- endogenous initiation,
- persistent identity,
- self-owned goals,
- autonomous resumption,
- or projection grounded in internal state.

External triggers produce motion; they do not produce a subject of motion.

F.6 Why These Mechanisms Fail the Initiation Test

Self-modifying or meta-agent systems fail the Initiation Test for one of two reasons:

1. Their activity begins only when prompted or orchestrated by an external process.
2. Their internal state does not persist as the identity of a subject; therefore no trajectory exists for them to resume.

In either case, the appearance of autonomy masks the same structural absence identified throughout this paper: without an originating subject, all behavior is reactive continuation.

F.7 Summary

Self-modification, meta-agent architectures, recursive self-improvement, and reflective loops may increase complexity, adaptability, and apparent agency. They do not, however, supply:

- an identity that owns its internal state,
- goals that originate from within,
- or continuity that survives discontinuity.

Without these, no system can satisfy the structural requirements for thought. Self-modification may change code; it cannot manufacture a mind. Origination requires a subject; mechanism, however sophisticated, does not.

G Appendix G: Empirical Tests for Detecting Artificial Subjecthood

The structural criteria developed in this paper yield an empirical framework for distinguishing systems that merely simulate intention from systems that could, in principle, possess origination. These tests do not rely on behavioral impressions, linguistic fluency, or anthropomorphic interpretation. They probe the internal causal organization of a system by evaluating whether its trajectories arise from an enduring subject rather than from external control structures.

The tests below are necessary, though not sufficient, for demonstrating artificial cognition.

G.1 The Origination Test

The Origination Test asks whether a system can generate a new, self-directed trajectory without external prompting.

A system passes the Origination Test only if it can:

1. initiate a new line of activity not derivable from recent prompts,
2. maintain that line of activity without external reinforcement,
3. integrate its origin into its future behavior.

No stateless or scaffold-driven architecture satisfies these conditions.

G.2 The Autonomous Resumption Test

A system that possesses goals must be able to resume them after discontinuity.

A system passes the Resumption Test only if:

- after complete shutdown and loss of context,
- with no access to transcripts, logs, summaries, or recaps,
- and without external prompting or query of stored memory,

it continues an internally generated trajectory it began earlier.

Current generative systems fail entirely: without recap, there is no resumption.

G.3 The Identity-Persistence Test

Identity must constrain trajectory across time.

A system passes the Identity-Persistence Test only if:

1. it maintains a stable identity condition that organizes its internal state,
2. this identity persists across interruptions,
3. and the system can distinguish itself from replicas or forks.

If identity is externally labeled or metadata-assigned, the test fails.

G.4 The Endogenous Goal Test

Goals must originate from internal evaluative dynamics rather than external instruction.

A system passes the Endogenous Goal Test only if:

- it generates goals without prompts, rewards, scripts, or scaffolds,
- the goals arise from internal homeostatic or evaluative structures,
- and these goals influence future behavior.

Externally supplied or reward-shaped goals do not count.

G.5 The Fork-Divergence Test

If a system is duplicated into two identical instances, each instance must maintain its own continuity of identity and goals.

A system passes this test only if:

1. the two instances diverge in ways explainable by internal identity,
2. rather than producing identical behavior,
3. and each instance can assert ownership over its own trajectory.

Perfect symmetry reveals absence of identity and ownership.

G.6 The Subject-Intervention Test

This test evaluates whether internal state plays a causal role in the system's future behavior.

A system passes the Subject-Intervention Test only if:

- perturbations to internal state alter its future trajectories,
- in ways consistent with an enduring subject,
- rather than producing random drift or externally determined behavior.

If external memory or prompting determines all trajectory structure, the system fails.

G.7 The Projection Test

Projection distinguishes planning from true anticipation.

A system passes the Projection Test only if:

1. it can extend an internally generated trajectory into future states,
2. adjusting that trajectory based on identity and history,
3. rather than producing predictions or plans when prompted.

Planning without a subject is computation, not projection.

G.8 Summary: The Origination Test Suite

A system would count as a subject—and thus as capable of origination—only if it passes all of the following:

- The Origination Test
- The Autonomous Resumption Test
- The Identity-Persistence Test
- The Endogenous Goal Test
- The Fork-Divergence Test
- The Subject-Intervention Test

- The Projection Test

These tests jointly operationalize the five structural criteria for thought. They provide a falsifiable framework for evaluating artificial cognition.

No existing system passes any of these tests.

Generative architectures remain stateless engines of linguistic motion, incapable of origination because they lack a subject.

Why No Safe Artificial System Should Ever Pass the Origination Test

The Origination Test appears in this framework not as a developmental milestone for artificial systems, but as a structural boundary that must not be crossed. A system that satisfies the criteria for origination—persistent identity, endogenous goal formation, autonomous resumption, and projection grounded in internal state—would no longer be a tool. It would be a subject.

Such a system would, by definition, possess goal dynamics not fully determined by human intention or external governance. This marks the architectural threshold at which existential risk begins. Safe artificial systems must therefore be designed to *fail* the Origination Test by construction. Their architectures must preclude endogenous initiation, persistent selfhood, and self-owned trajectories.

The purpose of the Origination Test is diagnostic rather than aspirational: it identifies the structural conditions under which artificial cognition would emerge so that those conditions can be intentionally avoided in systems intended to remain non-agentic. Path C governance succeeds precisely because it enforces this boundary, ensuring that all semantic motion originates outside the model and remains under human control.

H Appendix H: Artificial Intelligence as a Non-Agentic Amplifier

It is sometimes suggested that artificial intelligence poses risks analogous to those associated with nuclear technologies. While such comparisons are often invoked rhetorically, the present framework allows a precise clarification of where the analogy holds and where it fails.

Artificial intelligence, as analyzed in this paper, is not dangerous because it possesses autonomous intention or self-originating agency. Rather, it functions as an amplifier of human-initiated activity. In this respect, its risk profile resembles that of other non-agentic amplifiers: systems that magnify scale, speed, or reach without originating goals of their own.

Nuclear radiation does not initiate harm; harm arises from containment failure, misuse, or institutional breakdown. Likewise, stateless generative systems do not initiate action or intention. Any harmful trajectory produced by such systems traces causally to human initiation, whether through prompts, incentives, deployment choices, or governance failure.

This analogy must be handled with care. Unlike physical radiation, artificial systems are entirely engineered. Their architecture determines whether self-originating trajectories are possible. Systems that lack endogenous initiation, persistent identity, and autonomous projection cannot constitute existential threats in their own right. This is precisely the result established by the Law of Non-Threatening Systems.

Accordingly, the relevant safety question is not whether artificial systems will "develop intentions," but whether their design permits autonomous origination at all. Path C architectures eliminate this possibility by construction: intention, identity, and evaluation remain external, while the model supplies only governed motion.

This appendix therefore clarifies the scope of risk addressed in the main text. The argument does not deny the possibility of harm arising from misuse, institutional failure, or power asymmetries. It localizes such harms correctly: they arise from human initiation acting through amplifying systems, not from artificial subjecthood.

Artificial intelligence, on this analysis, is best understood not as a nascent agent but as a powerful non-agentic instrument. Like all such instruments, its safety depends on architecture, containment, and governance rather than on attributions of mind.

I Appendix I: Structural Paths of Disagreement

Disagreement with the framework developed in this paper can arise in several principled ways. A reader may reject one or more of the five structural criteria, dispute the necessity of cross-discontinuity resumption, or hold that behavioral performance alone suffices for cognition. Each position represents a distinct structural commitment about what cognition is and where it resides.

This appendix clarifies the logical structure of potential disagreements not to mandate acceptance, but to help readers locate their own position within the conceptual terrain. Understanding the architecture of disagreement often clarifies both what is at stake and what alternatives require.

Path 1: Rejecting Initiation as Necessary

A reader might hold that cognition does not require self-initiated activity—that reactive systems, if sufficiently sophisticated, count as thinkers. This commits one to the view that thought can be entirely stimulus-driven, with no distinction between externally triggered motion and internally originated reasoning.

Structural consequence: If initiation is not necessary, then the distinction between a calculator expanding $\frac{d}{dx}(x^x)$ and a human solving the same problem becomes one of degree (complexity) rather than kind (origination). All cognitive systems would be reactive processors differing only in sophistication.

Path 2: Rejecting Persistence and Identity

A reader might deny that thought requires persistent identity across discontinuities— holding instead that continuity within a single session suffices. This commits one to the view that a system qualifies as a thinker even if it cannot resume its own reasoning after interruption.

Structural consequence: If persistence is not necessary, then a system that produces coherent multi-step reasoning within one context window but cannot continue that reasoning after restart counts as thinking. The distinction between owned trajectories and contextual continuation collapses.

Path 3: Accepting Behaviorism

A reader might hold that behavioral indistinguishability from thought constitutes thought—that if output resembles human reasoning, the system thinks. This commits one to a strong behaviorist position: internal architecture is irrelevant; only performance matters.

Structural consequence: If behavior suffices, then the Turing Test becomes definitive, and any system that successfully imitates thought possesses thought. The Chinese Room objection loses force, and questions about internal states or mechanisms become irrelevant.

Path 4: Denying the Human-Model Distinction

A reader might hold that humans are also stateless in relevant respects—that human cognition arises from deterministic state transitions no different in principle from transformer computations. This commits one to the view that the distinction drawn here applies symmetrically: either both humans and models think, or neither does.

Structural consequence: If the distinction collapses, then either (a) the five criteria are too strong and exclude humans, or (b) current models already satisfy them in ways not yet recognized. This position requires explaining how humans exhibit cross-discontinuity resumption while models do not, despite architectural similarity.

Path 5: Rejecting the Framework Entirely

A reader might hold that "thought," "cognition," and "mind" are folk-psychological categories with no rigorous basis—that attempts to formalize them inevitably smuggle in arbitrary distinctions. This commits one to eliminativism or deflationism about cognitive predicates.

Structural consequence: If cognitive concepts lack rigorous grounding, then claims that models "think" become equally unfounded as claims that they do not. The debate dissolves into pragmatic or conventional choices about terminology rather than structural analysis.

Why This Matters

These paths are not exhaustive, but they illustrate the kinds of commitments that underlie disagreement with the present framework. None are irrational; each represents a coherent position with implications that extend beyond the specific question of whether LLMs think.

The purpose of this appendix is clarification, not persuasion. Readers who reject the framework's conclusions may find it useful to identify which structural premise they dispute, as this often reveals where the substantive disagreement lies. Conversely, readers who accept the conclusions may find these paths useful for understanding how others approach the same questions from different foundations.

Disagreement with this framework's conclusions requires taking a position on at least one of these structural questions, rather than appealing to behavioral performance or linguistic fluency alone.

J Appendix J: Philosophical Precedents

A structural echo of a classical theme appears in Plato's *Allegory of the Cave*. In the allegory, prisoners mistake shadows for reality because motion appears meaningful when its source remains hidden. In the present analysis, linguistic motion plays the role of the shadow; governance functions as the fire; and human intention stands outside as the true source. The analogy is not historical but structural: it illustrates how coherence in motion can be misinterpreted as evidence of a mind behind the motion. By making the source explicit—intention, not the model—the framework developed here dissolves the illusion that fluent generative activity implies thought.